

4. Consider the vectors $v_1 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$, $v_2 = \begin{pmatrix} 3 \\ 4 \\ 1 \end{pmatrix}$, $v_3 = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$ and let A be the matrix with these vectors as columns.

a) Is the matrix A singular?

b) Does A^{-1} exist?

c) Let $a = 1$, $b = 2$, $c = 3$ and find the linear combination $a v_1 + b v_2 + c v_3$

d) Are the rows of A linearly independent?